

Scanning Strategy Learning For Electronic Support Receivers by Robust Principal Component Analysis

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ABSTRACT

Narrow-band Electronic Support receivers cannot detect radar signals in broad frequency ranges of the electromagnetic spectrum simultaneously. Hence, a frequency spectrum scanning strategy has to be planned. Commonly, this strategy is determined based on prior knowledge about possible threats. However, in an environment where the parameters of the radars are unconfirmed, it could be planned via learning-based representations. In previous researches, this sensor scheduling problem was modeled as a dynamical system by Predictive State Representations. Moreover, Singular Value Thresholding (SVT) algorithm is used in the subspace identification part to cope with the complexity of the system. In this work, We propose a scanning strategy learning method based on Robust Principal Component Analysis (RPCA).

Keywords: Electronic Support Measures, Frequency Search Strategy, Robust Principal Component Analysis (RPCA), Scanning Regime Learning, Electronic Support Receiver, Electronic Warfare

1. INTRODUCTION

Receivers with high sensitivity but narrow bandwidth used in Electronic Support (ES), which is a sub-section of Electronic Warfare, have to scan a wide part of the electromagnetic spectrum over time in order to detect Radar signals operating in different frequency ranges. At this point, an optimization problem arises which consists of determining the best frequency spectrum search strategy for narrow bandwidth receivers. If this can be solved, an ESM system's detection performance will increase in whole covered electromagnetic spectrum not only in a specific frequency band. This scanning strategy can be established by deterministic calculations using previously obtained Radar signal parameters or by stochastic approaches. The performance of the strategies developed with deterministic methods does not increase over time, and they highly depend on the accuracy of the parameters obtained from other Electronic Intelligence (ELINT) systems. Therefore, in an environment where the parameters of Radar signals are not known, it would be more appropriate to use learning-based methods.

In previous deterministic approaches the spectrum search strategy is determined to achieve a better intercept probability (POI)^{1,2} or a better average intercept time.^{3,4} However, these strategies depend heavily on the accuracy of the prediction information. A small phase shift or latency can lead to significant performance degradation. This synchronization problem has been extensively studied.⁵ When there is no prior information about the environment, cognitive sampling strategies can be considered the best strategy.⁶ In recent years, dynamic planning strategies capable of learning from measured data have been studied^{7,8} These works have modelled this problem as a dynamically controllable environment.

In previous studies examining the problem of detecting the Radar signals by a narrow-band electronic support receiver with learning-based methods has been modeled as a dynamic system. In previous stochastic approaches, Transformed Predictive State Representations (TPSR)⁹ are used to predict future states of the system. With these estimated states, the frequency scanning strategy to be applied by the receiver can be determined. After, gathering each amount of data or observation about the environment, the strategy gets more accurate, and the number of signal interception increases. The optimization problem that arises with this approach was solved with iteratively applied singular value thresholding (SVT) or singular value decomposition (SVD) algorithms. In this work, we propose a new learning-based method to maximize total interception ratio of emitters by an ES receiver via Robust principal component analysis (RPCA).¹⁰ The emitters' illumination pattern and receiver's frequency search pattern are modeled as a dynamical system by using transformed predictive state representations (TPSR), likewise the previous approaches. Hence, the scanning strategy is learnt and updated over time. Unlike other methods, we used Robust Principal Component Analysis (RPCA) for subspace identification.

2. BACKGROUND ON SCANNING REGIME LEARNING

The radar interception from a narrowband electronic support receiver problem could be represented by a Predictive State Representation (PSR)¹¹ as a dynamic environment. The main purpose of representing a dynamical system with PSR is that if all the expected results of all possible action-observation pairs are recognised, the state of the system in a certain time can be calculated.¹¹ So, the prediction of the following state can be formulated as follows:

$$\mathbf{m}_{t+1} = \frac{M_{ao}\mathbf{m}_t}{\mathbf{m}_\infty^\top M_{ao}\mathbf{m}_t} \quad (1)$$

Where the system transition matrix is denoted as M_{ao} , current state vector at time t denoted as $\mathbf{m}_t$ and the normalization vector denoted as $\mathbf{m}_\infty$. According to (1), the next state of the system can be predicted $\mathbf{m}_{t+1}$, when the current state of the system $\mathbf{m}_t$ and M_{ao} are known.

In some systems, gathering exact values of the PSR parameters M_{ao} , $\mathbf{m}_t$ and $\mathbf{m}_\infty$ could not be possible. For these systems, a new technique is proposed to determine the estimation of PSR parameters based on the noticeable data rather than exact PSR parameters.⁹ These estimated PSR parameters represented with the three probability matrices P_H , $P_{T,H}$ and $P_{T,ao,H}$.⁹ The first probability matrix P_H holds the occurrence probability of each linearly independent action-observation pair in the history set denoted as H . History set constructed by collecting the past linearly independent action-observation pairs. Likewise, the second observable matrix $P_{T,H}$, holds the joint probabilities of the possible future action-observation pairs and past action-observation pairs from H . The final observable matrix is $P_{T,ao,H}$ carries the conditional probabilities of triples of the past, current and expected state for each possible action-observation pair. They also showed that expected values could be used rather than probability matrices when a uniformly distributed and sufficient number of data is gathered. Because measures will converge to true matrices according to the law of large numbers.

Despite the benefit of predicting the future status by PSR, the complexity of the description will expand after every repetition. Hence, a new representation model, Transformed PSR, is suggested to get a low dimensional representation that is related to actual PSR parameters.⁹ Transformed PSR (TPSR) parameters can be obtained by a linear transformation $J = U^\top R$ to PSR parameters. In recent works in this field, a new method is suggested to determine the transformation matrix J and gathering U matrix described above.⁷ In the subspace identification part, they generated a new probability matrix F and instead of directly applying the Singular Value Decomposition (SVD) algorithm to the estimated probability matrix $P_{T,H}$, they applied Singular Value thresholding (SVT) algorithm to F . The F matrix is given as below:

$$F = (P_{T,H}, P_{T,a_1o_1,H}, \dots, P_{T,a_{|A|}o_{|O|},H}) \quad (2)$$

where sub-indices $|A|$ and $|O|$ denotes the rank of the possible action and observation set, respectively. With this approach, a larger subspace is obtained which helps to avoid large prediction errors due to small subspace approximation and also noise removed from the system dynamics via SVT. Moreover, Nuclear norm minimization is applied to F as below formulation:

$$\min_{F \in \mathbb{R}^{n \times m}} \mu \|F\|_* + \frac{1}{2} \|F - \hat{F}\|^2 \quad (3)$$

So that, a new matrix obtained $\tilde{F}$ and SVD_l applied to $\tilde{F}$ to get a new $\tilde{U}$ where l denotes the rank given from SVT. Moreover, $P_{T,H}$ and $P_{T,ao,H}$ extracted from $\tilde{F}$.

By applying that new method, subspace identification is applied to a larger set. This may point to evading severe estimation inaccuracies if the $P_{T,H}$ matrix is a noisy matrix. With this approach, some of the system transition matrix dimensions are modelled as noise to get a more steady probability matrix. After the subspace identification part, the estimation of the next state could be done. The association among the probability matrices and the TPSR parameters⁷ are given as follows.

$$\mathbf{m}_\star = U^\top P_{T,H} \mathbf{1} \quad (4)$$

$$\mathbf{m}_\infty^\top := P_H^\top (U^\top P_{T,H})^\dagger \quad (5)$$

$$M_{ao} := U^\top P_{T,ao,H} (U^\top P_{T,H})^\dagger \quad (6)$$

where, $^\top$ indicates the transpose of a matrix and † indicates the Moore-pseudo inverse of a matrix. Eventually, practical calculations can be done through using eq (1) to (6).

3. THE PROPOSED METHOD

In⁷ SVT practised for the subspace identification and the noise removal of the system dynamics. In this work, in contrast to the previous approaches, we applied Robust Principal Component Analysis (RPCA)¹⁰ to F matrix.

3.1 Subspace Identification via RPCA

Robust Principal Component Analysis is an algorithm suggested to cope with high-dimensional and highly corrupted images. This prominent technique gives low-rank and sparse decomposition that can be used in numerous applications such as video surveillance, face recognition and image denoising.¹² RPCA is applied for dimension reduction by decomposing highly corrupted data matrix into low-rank matrix and sparse matrix. Essentially, the mathematical basis behind RPCA starts with a matrix and its decomposition into two components. In our proposed method, F matrix can be decomposed as below:

$$F = L + S. \quad (7)$$

where L is the low-rank matrix and S is the sparse matrix. To obtain the low-rank and sparse components, a convex relaxation-based algorithm¹⁰ is proposed, assuming that $\|L\|_* := \sum_i \sigma_i(F)$ represents the nuclear norm of the input matrix F and $\|S\|_1 := \sum_{ij} |F_{ij}|$ specifies the L1-norm of the matrix F . With using these minimal assumptions, the approximation yields a solution for below formula:

$$\min \|L\|_* + \lambda \|S\|_1 \quad (8)$$

Hence, lowest-rank matrix L and sparse component S will be generated. In this convex problem, $\lambda > 0$ provides a balance between sparsity and rank. Moreover, the L1-norm of the matrix S enforces sparsity in S , and nuclear norm $\|L\|_*$ enforces low rank in L . This method is widely used due to the high probability of finding the exact components of a large matrix. The advantage of this approximation is that it is widely used under numerous conditions and for various data types.

The analytical model underlying that method relies on the augmented Lagrangian,

$$l(L, S, Y) = \|L\|_* + \lambda \|S\|_1 + \langle Y, F - L - S \rangle + \frac{\sigma}{2} \|F - L - S\|_F^2 \quad (9)$$

where Y is an $n \times r$ matrix whose entries are taken independently from a $N(0, 1/n)$ distribution and λ, σ are the Lagrange multipliers. The solution to eq (9) is obtained by minimizing l with respect to L and S , fixing one value at a time. This optimization can be done as below:

$$\underset{S}{\operatorname{argmin}} (l(L, S, Y)) = S_{\lambda\sigma}(F - L + \sigma^{-1}Y) \quad (10)$$

$$\underset{L}{\operatorname{argmin}} (l(L, S, Y)) = D_\sigma(F - S - \sigma^{-1}Y) \quad (11)$$

$S_\tau(1) = \operatorname{sgn}(x) \max(|x| - \tau, 0)$ in eq (10) is the shrinkage operator and $D_\tau(X) = USV^*$ in eq (11) is the singular value threshold operator, and $X = USV^*$ is an arbitrary singular value decomposition (SVD). The Augmented Lagrange -multiplier technique gives the best results considering the robustness and scalability of the data.¹⁰

3.2 State Prediction Update Policy

In this part, we introduce how we generate and update the scan regime over time. After subspace identification, the frequency band selections to be used in the following scan can be determined according to eq (1). In this phase, we implemented a stochastic random strategy used in multi-arm bandit problems to maximize the overall capture rate of radar signals. The formulation¹³ given below was used to select predicted activities.

$$P(\tilde{A}_t = k) = \frac{(1 - \sigma) \exp(\frac{\sigma}{\tau} \sum_{m \in b_k} P(m)_t)}{\exp(\sum_{m \in b_k} P(m)_t)} \quad (12)$$

This approach connects uniform random exploration and exponentially weighted exploitation to maximize the reward function.

4. SIMULATION SETUP AND RESULTS

We modelled the radar illumination patterns as cyclic pulse trains with arbitrary pulse width and arbitrary period. We formed different binary pulse trains per radar. Additionally, we assumed that each radar operates in a different frequency band. In experiments, we produce K possible frequency bands to scan. Consequently, there are K different possible actions a_t that can be performed by the receiver in time step t . We consider just one receiver scanning the frequency spectrum. Thus, there is only one agent that can discover and learn.

According to that model, if the radar pulse train is high at time t and the receiver is tuned to that frequency band at the same time, then the interception of radar illumination is supposed to be 1 at that time step. If the transmit signal is not activated, the amplitude of the received signal is weaker than the sensitivity of the receiver or the transmit signal does not point in the same direction as the receiver, then the radar illumination is confirmed to be 0 at that time step. Thus, there are only two potential observations, namely the negative observation $o_t = 0$ and the positive observation $o_t = 1$.

$$o_t = \begin{cases} 1, & \text{if the radar pulse train is positive at} \\ & \text{time } t \text{ and the chosen frequency band} \\ & \text{includes the radar frequency} \\ 0, & \text{otherwise.} \end{cases}$$

This parametric design ensures that the suggested approach can be applied to receivers with different detection bandwidths or different minimum detection times. In the practices, we pretended that the detection bandwidth of the receiver is 1 GHz, the smallest detection time is 100 ms, and the complete frequency coverage ranges from 2 GHz to 12 GHz.

In execution, the receiver begins with a simple periodic search strategy.¹ That is, the tuned frequency band of the receiver changes incrementally from 1 to K after each time step and repeats this model regularly until the first prediction is made. To obtain adequate statistics on the core test set, we did not modify the initial policy for k times of the prediction phase. This part of the process fixes the serious errors due to sampling. We updated the test and history sets after each measurement. The length of our test and history sets is L . We adjust our procedure according to the last L action observation pair.

In the experiments, we assumed that there are 10 possible frequency bands to search. Also, we modelled the radar signal as a periodic binary pulse train with a random period ranging from 40 to 80-time steps and arbitrary pulse width ranging from 10 to 20-time steps. In addition, the period and pulse width change at each illumination with 5% jitter to pretend the loss of detection of the beginning and end of illumination. Moreover, in this experiment, it was assumed that the effect of detection of each radar signal is the same. Therefore, the weight of each event is assumed to be fixed. From the start to the 300th timestamp, the frequency bands are selected by a simple search strategy. After this period, the prediction is performed for the next 30 states. The strategy is updated for every 300th-time step after initialization. The radar illuminations are reproduced for 30×10^4 time steps. Therefore, the total number of intercepted illuminations is projected for 2000 times.

The comparison of the singular values of F and L for this experiment's data-set is shown in Fig. 1.

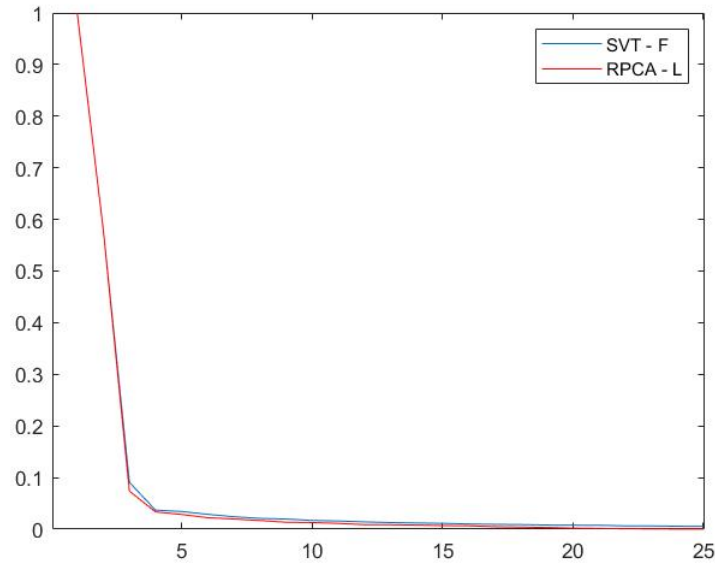


Figure 1. Singular Values of F and L

Fig. 1 shows that for a given data matrix F , the singular values of the recovered low-rank matrices $\hat{F}$ via the SVT algorithm and L via the RPCA are almost the same. Therefore, in the conventional method⁷ SVT can be trivially replaced by the RPCA algorithm and the desired PSR parameters can be estimated via eq (1) to (6). Fig. 2 shows the percentage of the total amount of intercepted illuminations on the total amount of illuminations.

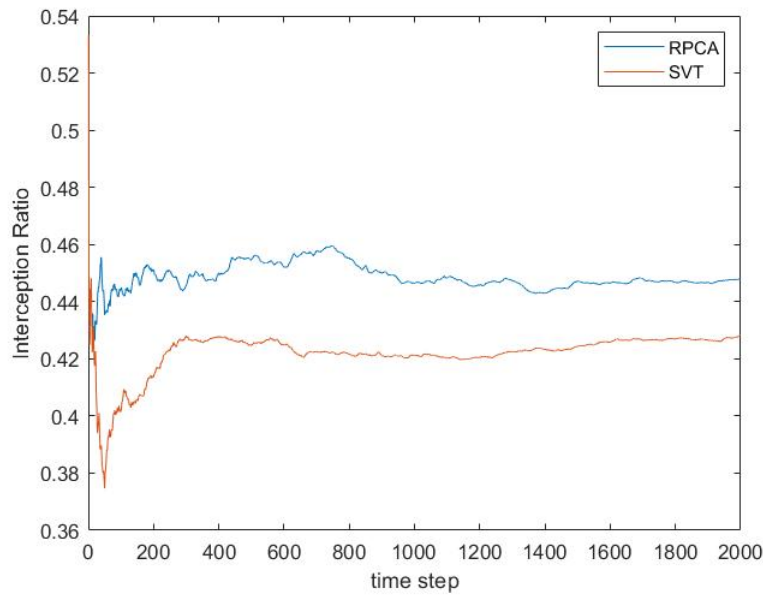


Figure 2. Signal Interception Ratios of RPCA and SVT based Methods

The red dotted line and the blue dotted line show the total intercepted illuminations by SVT and RPCA algorithm, respectively. In the simulations, the singular values are calculated for the values above 10^{-5} for both method. This figure shows that the performances of RPCA and SVT based methods are quite equal and the

performance increases slightly with time for both learning-based models. Moreover, the average intercept ratios of each frequency band in these simulations are given in Table 1. According to Table 1, it can be said that both

Table 1. Interception Ratios of Each Frequency Band

Frequency Band	Interception Ratio	
	RPCA	SVT
2-3 GHz	45,83%	46,24%
3-4 GHz	37,17%	39,27%
4-5 GHz	27,36%	26,29%
5-6 GHz	38,47%	35,43%
6-7 GHz	39,96%	40,35%
7-8 GHz	39,64%	40,41%
8-9 GHz	31,87%	31,36%
9-10 GHz	54,73%	51,07%
10-11 GHz	43,45%	41,09%
11-12 GHz	61,23%	58,51%

strategies did not get stuck at a particular local maximum point and accurately sampled the environment at all frequencies. Moreover, both strategies reached an adequate level of success on all bands.

5. CONCLUSION AND FUTURE WORK

In this paper, we modelled the problem of intercepting radar signals with an ES receiver as a dynamic environment. We mainly focused on the cognitive methods to update the sampling strategy over time. We proposed a new method for learning the sampling strategy by combining the recently developed matrix decomposition and predictive state representation algorithms. We used RPCA instead of SVT. In the future, different subspace identification techniques or matrix decomposition algorithms can be used to speed up the computations or develop a better prediction update policy. Moreover, the performance of the proposed method can be evaluated using a real ES support system instead of a simulation environment. The current research focuses on speed up the computational time consumption at subspace identification part.

REFERENCES

- [1] Köksal, E., *Periodic search strategies for electronic countermeasure receivers with desired probability of intercept for each frequency band*, Master's thesis (2010).
- [2] Winsor, C. and Hughes, E., "Optimisation and evaluation of receiver search strategies for electronic support," *IET Radar, Sonar & Navigation* **6**(4), 233–240 (2012).
- [3] Clarkson, I. V. L., El-Mahassni, E. D., and Howard, S. D., "Sensor scheduling in electronic support using markov chains," *IEE Proceedings-Radar, Sonar and Navigation* **153**(4), 325–332 (2006).
- [4] Clarkson, I. V. L., "Synchronisation in scan-on-scan problems," *Proceedings of the Defence Applications of Signal Processing (DASP'09)* (2009).
- [5] Kelly, S., Noone, G., and Perkins, J., "Synchronization effects on probability of pulse train interception," *IEEE transactions on aerospace and electronic systems* **32**(1), 213–220 (1996).
- [6] Xun, Y., Kokar, M. M., and Baclawski, K., "Using a task-specific qos for controlling sensing requests and scheduling," in *[Third IEEE International Symposium on Network Computing and Applications, 2004.(NCA 2004). Proceedings.]*, 269–276, IEEE (2004).
- [7] Claude, H., Enderli, C., Grandin, J.-F., and Pietquin, O., "Learning of scanning strategies for electronic support using predictive state representations," in *[2015 IEEE 25th International Workshop on Machine Learning for Signal Processing (MLSP)]*, 1–6, IEEE (2015).

- [8] Clarkson, I. V. L., “Intercepting beam-agile radar,” in [*2018 International Conference on Radar (RADAR)*], 1–6, IEEE (2018).
- [9] Boots, B., Siddiqi, S. M., and Gordon, G. J., “Closing the learning-planning loop with predictive state representations,” *The International Journal of Robotics Research* **30**(7), 954–966 (2011).
- [10] Candès, E. J., Li, X., Ma, Y., and Wright, J., “Robust principal component analysis?,” *Journal of the ACM (JACM)* **58**(3), 1–37 (2011).
- [11] Littman, M. L., Sutton, R. S., and Singh, S. P., “Predictive representations of state,” in [*NIPS*], **14**(1555), 30 (2001).
- [12] Bouwmans, T., Javed, S., Zhang, H., Lin, Z., and Otazo, R., “On the applications of robust pca in image and video processing,” *Proceedings of the IEEE* **106**(8), 1427–1457 (2018).
- [13] Auer, P., Cesa-Bianchi, N., Freund, Y., and Schapire, R. E., “The nonstochastic multiarmed bandit problem,” *SIAM journal on computing* **32**(1), 48–77 (2002).